

Jehyeon Shin

Graduate School of Artificial Intelligence
Pohang University of Science and Technology (POSTECH)
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Education

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|-------------------|---|
| 2023.03 - Present | Ph.D. in Graduate School of Artificial Intelligence, Pohang University of Science and Technology (Korea)
<i>Advisor: Junsuk Rho</i> |
| 2026.05 - 2027.02 | Visiting Scholar in Physics, Massachusetts Institute of Technology (USA)
<i>Advisor: Marin Soljačić</i> |
| 2024.10 - 2025.02 | Visiting Scholar in Electrical and Computer Engineering, Seoul National University (Korea)
<i>Advisor: Sunkyu Yu</i> |
| 2016.03 - 2023.02 | B.S. in Biosystems engineering, (Double major: Electrical and Computer Engineering), Seoul National University (Korea)
<i>Advisor: Joongyong Rhee</i> |

Skills

- [1] Programming Languages
Python, C/C++, Java
- [2] Machine Learning Frameworks
PyTorch
- [3] EM Wave Simulation Software
FDTD (Lumerical, Meep), FEM (COMSOL multiphysics), Geometric Optics (Zemax), RCWA

Awards & Honors

- [1] **2026 Overseas Dispatch Support Program for Graduate Students**, Pohang University of Science and Technology, Korea, 2026 - 2027 (KRW 20,000,000 / 10 months)
- [2] **Hyundai Motor Chung Mong-Koo Scholarship**, Hyundai Motor Chung Mong-Koo Foundation, Korea, 2025 - 2027
- [3] **NRF Ph.D. Fellowship**, National Research Foundation, Korea, 2025 - 2027
- [4] **Graduated *summa cum laude***, Seoul National University, Korea, 2023
- [5] **3rd place in 2022 SNU FastMRI Challenge**, Seoul National University, Korea, 2022

- [6] **Army Commendation Medal in Eighth United States Army**, U.S. Army, USA, 2021
- [7] **1st place in Deep Learning unmanned sales stand product recognition AI competition**, Seoul National University, Korea, 2021
- [8] **National Science & Technology Full-ride Scholarship**, Korea Student Aid Foundation, Korea, 2016 - 2022

Publications

* for mark for 1st author

- [1] Eunseo Cho*, Xiaotong Li*, Shiqi Hu*, Hongyoon Kim, **Jehyeon Shin**, Junsuk Rho, *Current Opinion in Solid State and Materials Science* **43**, 101262, 2026.0 (doi.org/10.1016/j.cossms.2026.101262) [IF: 13.4] (JCR 10% in Physics, Applied/Physics, Condensed Matter/Materials Science, Multidisciplinary)
- [2] Beomseok Oh*, Dongwoo Lee, Yeon-Seong Choo, Sung-Hoon Byun, **Jehyeon Shin**, SeaMoon Kim, Junsuk Rho, *Journal of Sound and Vibration*, 119919, 2026.0 (doi: 10.1016/j.jsv.2026.119919) [IF: 0]
- [3] Junkyeong Park*, Younghwan Yang, Yujin Park, Hyunjung Kang, **Jehyeon Shin**, Harit Keawmuang, WonSik Kim, Trevon Badloe, YoungKi Kim, Junsuk Rho, *Advanced Optical Materials*, 2403413, 2025.0 (doi: 10.1002/adom.202403413) [IF: 8] (JCR 10% in Optics)
- [4] Minseok Choi*, Junkyeong Park*, **Jehyeon Shin***, Harit Keawmuang*, Hongyoon Kim, Jooyeong Yun, Junhwa Seong, Junsuk Rho, *npj Nanophotonics* **1**, 31, 2024.0 (doi: 10.1038/s44310-024-00029-2) [IF: Pending]
- [5] Jeonghoon Park*, Jaebum Noh*, **Jehyeon Shin***, Grace X Gu, Junsuk Rho, *Advanced Functional Materials* **35**, 2412901, 2024.0 (doi: 10.1002/adfm.202412901) [IF: 18.5] (JCR 5% in Chemistry, Multidisciplinary/Chemistry, Physical/Materials Science, Multidisciplinary/Physics, Applied/Physics, Condensed Matter)
- [6] Niloufar Raeis-Hosseini*, Jaebum Noh, **Jehyeon Shin**, Junsuk Rho, *ACS Applied Electronic Materials* **6**, 3501-3509, 2024.0 (doi: 10.1021/acsaelm.4c00285) [IF: 4.3]
- [7] Shutao Zhang*, Cheng-Feng Pan*, Yandong Fan, **Jehyeon Shin**, Yuanda Liu, Yan Liu, Jun Ding, Jing Wu, Junsuk Rho, Yuri Kivshar, Joel K. W. Yang, Zhaogang Dong, *arXiv preprint*, (doi.org/10.48550/arXiv.2605.20605)

Conferences

- [1] SES Annual Technical Meeting, Atlanta, USA, October, 2025
- [2] Future Challenges for Nanophotonic Scattering, Bad Herrenalb, Germany, October, 2025
- [3] Metamaterials, Crete, Greece, September, 2024

Patents

Domestic Pending

- [1] **Jehyeon Shin**, Jeonghoon Park, Jaebum Noh, Junsuk Rho, generative model for design of metamaterials, metamaterial design method, and computer-readable storage medium for storing data to drive the generative model, Application number: 10-2024-0183150, (submission date: 2024.12.10), KR Patent
- [2] **Jehyeon Shin**, Jaebum Noh, Junsuk Rho, dissolvable biomemristor and manufacturing methods thereof, Application number: 10-2024-0161466, (submission date: 2024.11.13), KR Patent

Press & Broadcast

- [1] "Investigating Static and Dynamic Behaviors in 3D Chiral Mechanical Metamaterials by Disentangled Generative Models", [Donga Science] [E-science] [Kyosu News] [The Electronic Times] (pressed on 2025.01)

Peer Review

- [1] Advanced Photonics NEXUS (1 reviews, verified by ORCID)
- [2] npj Nanophotonics (1 reviews, verified by ORCID)
- [3] Optics express (1 reviews, verified by ORCID)

Teaching Assistant

- [1] Deep Learning, AIGS538, 2025-1
- [2] Object-Oriented Programming, CSED232, 2024-1